DESIGN AND IMPLEMENTATION OF A DBMS FOR RECRUITMENT AND APPLICANT TRACKING SYSTEM

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1.PROBLEM IDENTIFICATION

- Recruitment information is often scattered across emails, spreadsheets and separate records.
- It becomes difficult to maintain a complete history of each candidate.
- Manual tracking can lead to duplicate data, missed interviews and inconsistent application statuses.
- HR departments need a centralized system to manage:
  -Vacancies
  -Candidates
  -Applications
  -Interviews
  -Evaluations
  -Offers
  -Joining records
- Therefore, we propose a relational database system that connects the complete recruitment lifecycle

2.SCOPE OF THE SYSTEM

Vacancy management

- Create and manage job vacancies
- Define required positions and jobs
- Track vacancy status

Candidate Management

•Store candidate information.
•Maintain skills and qualifications.
•Keep candidate records organized.

Application Tracking

•Record applications for vacancies.
•Track application status.
•Manage shortlisting and selection.

Interview Management

- Create interview rounds.
- Assign interview panels.
- Schedule interviews.
- Record evaluations.

Offer & Joining

- Generate offers for selected candidates.
- Track offer status.
- Record joining information.

Reporting

- Generate recruitment reports.
- Track vacancy progress.
- Analyze applications, interviews and hiring.

3.PROJECT OBJECTIVES

- **Centralize Recruitment Data**
  To store all recruitment related data under one database.
- **Reduce Data Redundancy**
  To normalize the data upto the Third Formal Form (3NF)
- **Maintain Data Integrity**
  Use Primary keys, Foreign keys, UNIQUE constraints, NOT NULL constraints and DEFAULT values
- **Track Candidate Progress**
  Follow candidates throughout the whole process(ie-application>interview>evaluation>offer>joining)
- **Support Reporting and Analysis**
  Use SQL queries, joins, aggregate functions, nested queries and views
- **Provide a Foundation for a Front End**
  Develop a small application using typescript

4.USERS OF THE SYSTEM

HR / Recruiter

- Creating and managing vacancies
- Reviewing applications
- Shortlisting candidates
- Scheduling interviews
- Recording evaluations
- Issuing offers
- Tracking joining status

Interview Panel

- View assigned interviews
- Conduct interview rounds
- Evaluate candidates
- Enter scores and remarks
- Provide recommendations

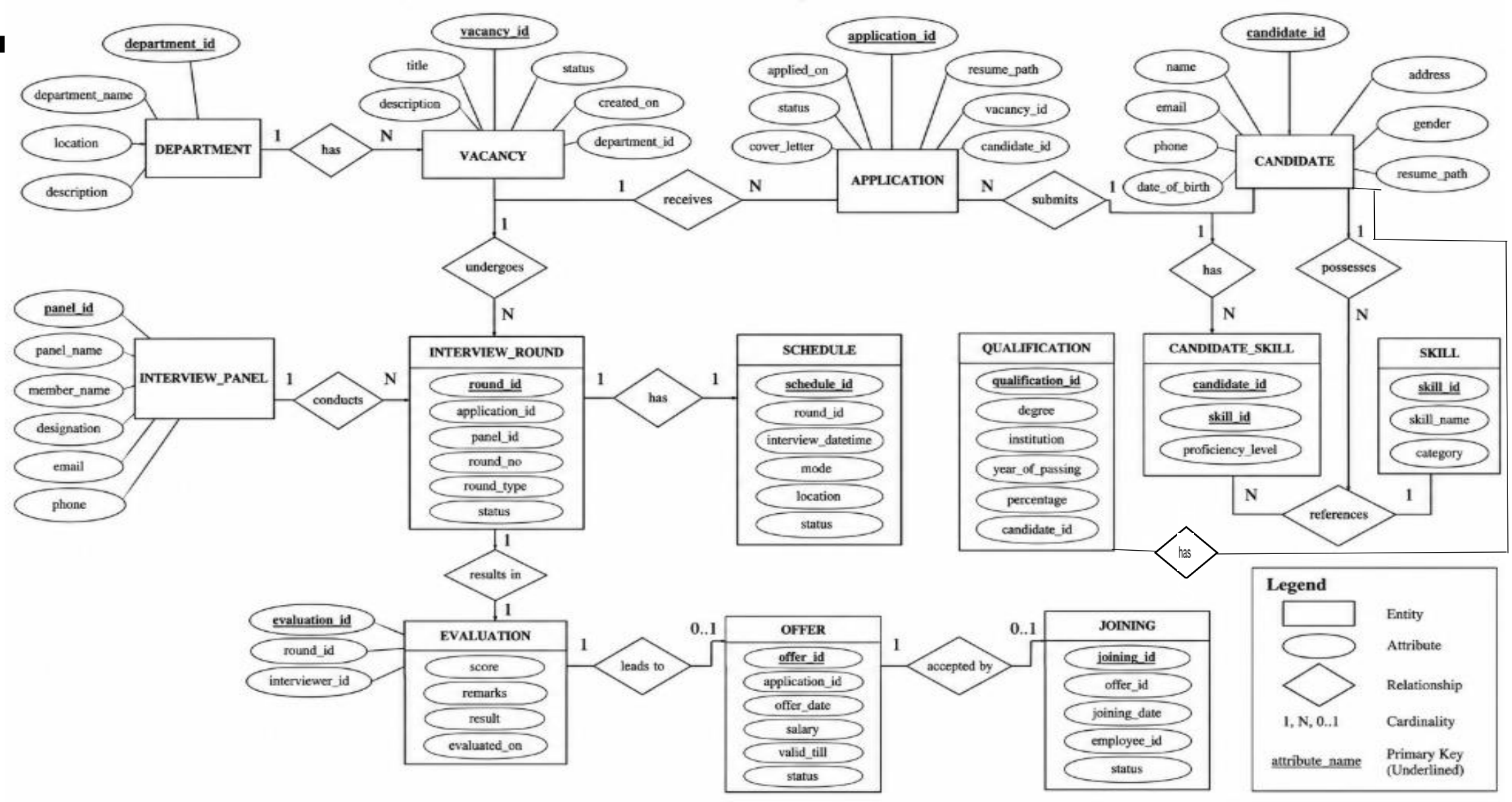
Administrator / Management

•Monitor recruitment progress
•View vacancy status
•Review offers and joining records
•Generate recruitment reports
•Maintain important system data

5.FUNCTIONAL REQUIREMENTS

- **Create and manage** job vacancies.
- **Register** candidates with their personal information.
- **Store** candidate skills and qualifications.
- **Create and track** applications.
- **Shortlist** candidates for suitable vacancies.
- **Create and manage** interview rounds.
- **Assign** interview panels.
- **Schedule** interviews with appropriate dates and times.
- **Record** evaluation scores, remarks and results.
- **Create** offers for selected candidates.
- **Record** the candidate's joining information.
- **Generate** useful reports such as:
  -Applicant reports
  -Shortlisted candidates
  -Interview schedules
  -Offer status
  -Vacancy progress

6.ER DIAGRAM


department_id
department_name
location
description
DEPARTMENT
1
has
N
vacancy_id
title
description
status
created_on
department_id
VACANCY
1
receives
N
application_id
applied_on
status
cover_letter
resume_path
vacancy_id
candidate_id
APPLICATION
N
submits
1
candidate_id
name
email
phone
date_of_birth
address
gender
resume_path
CANDIDATE
1
undergoes
N
1
has
N
1
possesses
N
panel_id
panel_name
member_name
designation
email
phone
INTERVIEW_PANEL
1
conducts
N
INTERVIEW_ROUND
round_id
application_id
panel_id
round_no
round_type
status
1
has
1
SCHEDULE
schedule_id
round_id
interview_datetime
mode
location
status
QUALIFICATION
qualification_id
degree
institution
year_of_passing
percentage
candidate_id
CANDIDATE_SKILL
candidate_id
skill_id
proficiency_level
SKILL
skill_id
skill_name
category
N
references
1
has
1
results in
1
evaluation_id
round_id
interviewer_id
EVALUATION
score
remarks
result
evaluated_on
1
leads to
0..1
OFFER
offer_id
application_id
offer_date
salary
valid_till
status
1
accepted by
0..1
JOINING
joining_id
offer_id
joining_date
employee_id
status
Legend
Entity
Attribute
Relationship
1, N, 0..1
Cardinality
attribute_name
Primary Key
(Underlined)

WEEK 1 DELIVERABLES

- Problem Identification
- Scope
- Objectives
- Users
- Functional Requirements
- Conceptual design
- ER Components and Diagram

WEEK 2 (NEXT WEEK) DELIVERABLES

- Database Normalization
- Define Constraints
- Add Business Rules
- Populate Sample Data
- Develop SQL Queries
- Develop Front End
- Testing

THANK YOU